

WEEK 4 – POWER BI EDA REPORT

SHOPIFY STOCK ANALYSIS

1. PROJECT OVERVIEW :

Introduction :

During Week 4, I worked on **Exploratory Data Analysis (EDA) of Shopify stock data using Microsoft Power BI**. The main purpose of the analysis was to understand stock price movements, trading volume, and moving-average trends over time.

The dashboard was designed with interactive features such as date filters, dynamic price selection, comparison charts, and a custom tooltip. These features make it easier to explore the stock data from different perspectives.

Key Areas Covered :

- Stock Price Trend Analysis
- Trading Volume Analysis
- 20-Day Simple Moving Average
- 50-Day Simple Moving Average
- Moving Average Crossover
- Date Range Filtering
- Relative Date Filtering
- Dynamic Price Metric Selection
- High and Low Price Comparison
- Open and Close Price Comparison
- Custom Tooltip
- Interactive Power BI Dashboard

2. DATA AND ANALYSIS PREPARATION :

The Shopify stock dataset contains time-based stock information that can be used to study price and trading activity.

The analysis focused on different stock price fields such as:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

These fields were used to create different measures and visualizations for understanding stock behaviour.

The main analytical focus was on comparing actual values with moving-average values and exploring changes across different time periods.

3. BUILDING THE MOVING AVERAGE LAYER:

3.1 20-Day Simple Moving Average :

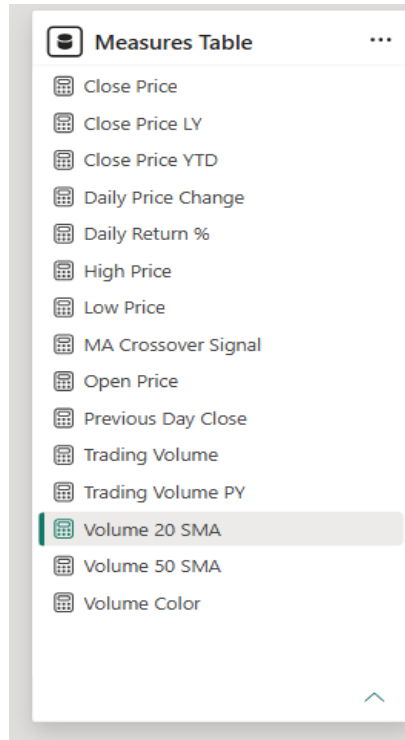
A **20-Day Simple Moving Average (20 SMA)** was created to calculate the average Shopify trading volume over a 20-day period.

The measure uses `DATESINPERIOD()` to define the required time period and `AVERAGE()` to calculate the average volume.

DAX Measure

```
Volume 20 SMA = VAR CurrentDate = MAX('Date  
Table'[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod = DATESINPERIOD( 'Date Table'[Date],  
CurrentDate, -PeriodDays, DAY )  
RETURN CALCULATE( AVERAGE('SHOP_2015-05-21_2025-03-  
16'[Volume]), DateInPeriod )
```

OUTPUT: A 20-day average-volume measure was created successfully.



4. CREATING THE 50-DAY MOVING AVERAGE

4.1 50-Day Simple Moving Average :

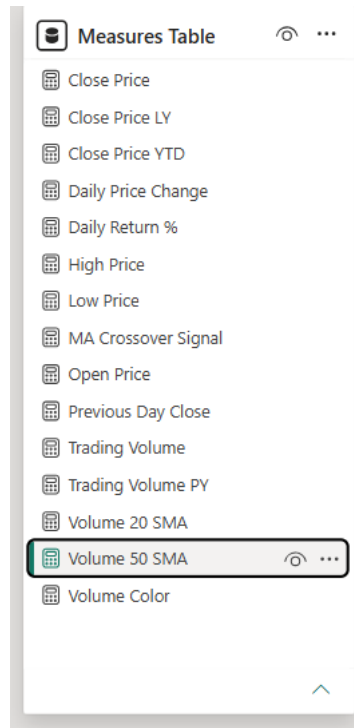
A **50-Day Simple Moving Average (50 SMA)** was created to provide a longer-period reference for trading volume.

The same DAX approach used for the 20-day SMA was applied, but the calculation period was changed to 50 days.

DAX Measure :

```
Volume 50 SMA = VAR CurrentDate = MAX('Date
Table'[Date])
VAR PeriodDays = 50
VAR DateInPeriod = DATESINPERIOD( 'Date Table'[Date],
CurrentDate, -PeriodDays, DAY )

RETURN CALCULATE( AVERAGE('SHOP_2015-05-21_2025-03-
16'[Volume]), DateInPeriod )
```



5. ***MOVING AVERAGE CROSSOVER SIGNAL:***

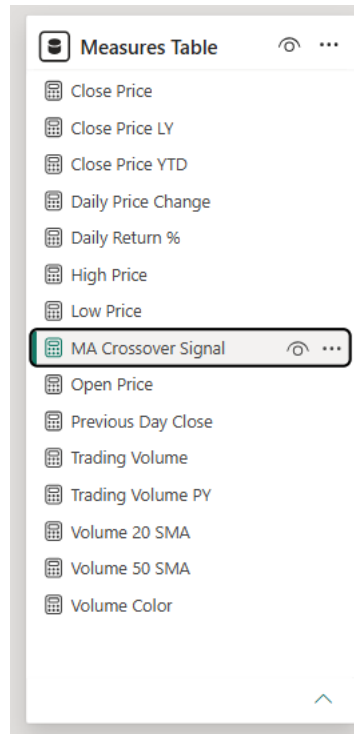
5.1 ***Crossover Analysis :***

A **Moving Average Crossover Signal** was created by comparing the 20-day SMA with the 50-day SMA.

The purpose of this measure was to identify the relationship between the shorter-term and longer-term moving averages.

DAX Measure :

```
MA Crossover Signal = IF( ISBLANK([Volume 20 SMA]) ||
ISBLANK([Volume 50 SMA]), BLANK(), IF([Volume 20 SMA] >
[Volume 50 SMA], "Bullish (20 > 50)", "Bearish (20 <
50) ") )
```



6. READING THE PRICE MOVEMENT :

6.1 Price Trend and Moving Averages :

A main **Price Trend & Moving Averages** chart was created to study the relationship between Shopify's closing price and the moving-average trends.

Visual Configuration

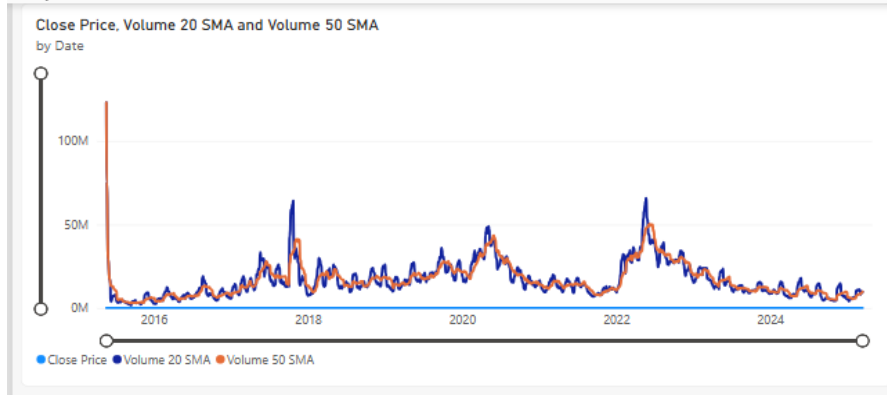
X-Axis: Date

Values:

- Close Price
- 20 SMA
- 50 SMA

The date axis was configured as continuous, and a **Zoom Slider** was enabled.

This allows the stock price and moving averages to be examined across the timeline while also making it easier to focus on a particular section of the data.



7. *ANALYSING TRADING ACTIVITY :*

7.1 *Trading Volume and 20-Day SMA :*

A separate chart was created to compare the actual trading volume with its 20-day moving average.

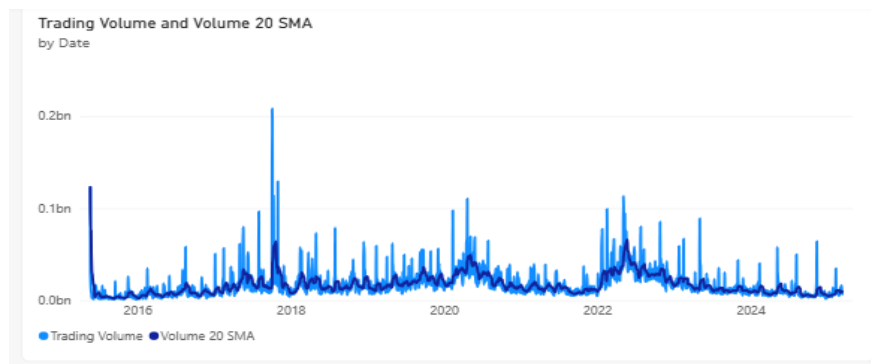
Visual Configuration

X-Axis: Date

Values:

- Trading Volume
- Volume 20 SMA

The visualization provides a comparison between daily trading activity and its recent average. This makes changes in trading activity easier to observe over time.



8. MAKING THE DASHBOARD INTERACTIVE:

8.1 Date Range Slicer :

A **Date Range Slicer** was added using the date field.

The slicer was configured in **Between mode**, allowing a starting date and ending date to be selected.

This enables the analysis to focus on a specific period of Shopify stock data.



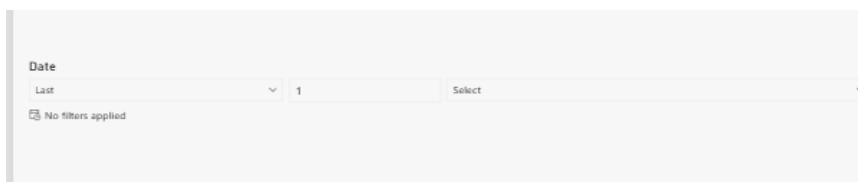
8.2 Relative Date Slicer :

A **Relative Date Slicer** was added to provide quick access to different time periods.

The available selections include:

- 1 Month
- 3 Months
- 6 Months
- YTD
- 1 Year

This feature provides a faster way to change the analysis period without manually selecting dates.



9. DYNAMIC PRICE METRIC SELECTION :

9.1 Metric Selector :

A **Metric Selector** was created using a Field Parameter.

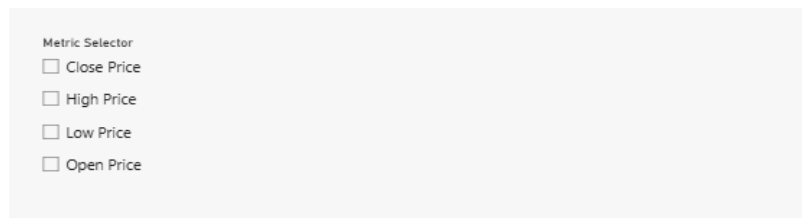
The selector includes four different price measures:

- Close Price
- High Price
- Low Price
- Open Price

The field parameter was created through:

Modeling → New Parameter → Fields

This allows the user to switch between different stock price measures interactively without creating separate dashboards for each measure.



10. COMPARING PRICE BEHAVIOUR :

10.1 High and Low Price Trend :

A line chart was created to compare the **High Price** and **Low Price** of Shopify stock across different dates.

Visual Configuration

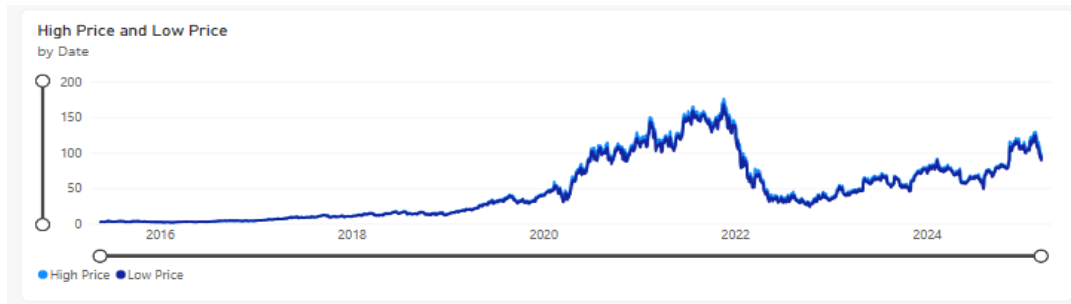
X-Axis: Date

Values:

- High Price
- Low Price

Chart Type: Line Chart

This visualization provides a direct comparison of the highest and lowest stock prices over the selected period.



10.2 Open and Close Price Trend :

Another line chart was created to compare the **Open Price** and **Close Price** over time.

Visual Configuration

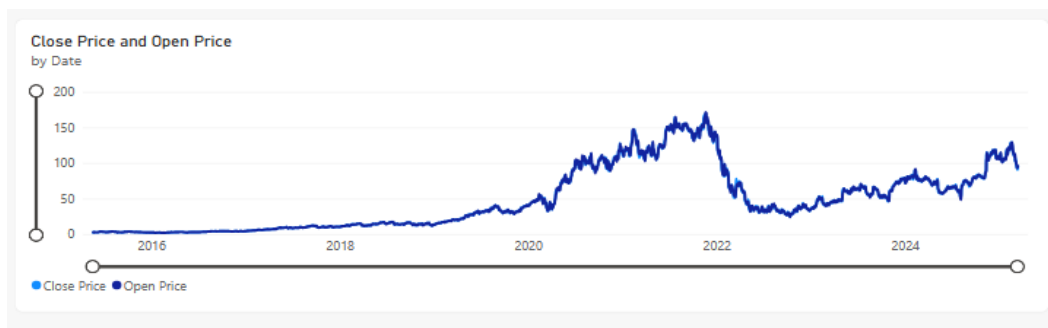
X-Axis: Date

Values:

- Open Price
- Close Price

Chart Type: Line Chart

This comparison helps in observing how the opening and closing prices changed throughout the stock's trading history.



11. CUSTOM TOOLTIP :

11.1 Stock Tooltip :

A separate report page named **Stock_Tooltip** was created to provide additional information while interacting with the main price chart.

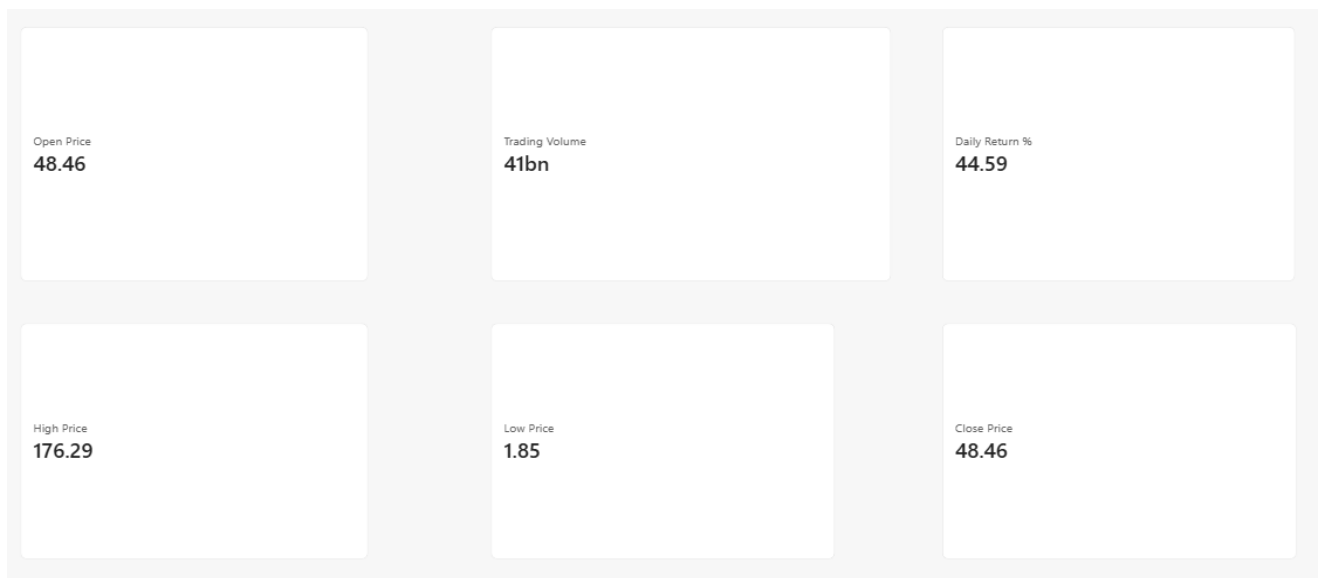
The page was configured as a tooltip by enabling **Allow as tooltip** and setting the canvas type to **Tooltip**.

Information Included

The tooltip displays:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Daily Change %

The tooltip was connected to the main price chart so that additional stock information could be viewed while interacting with the visual.



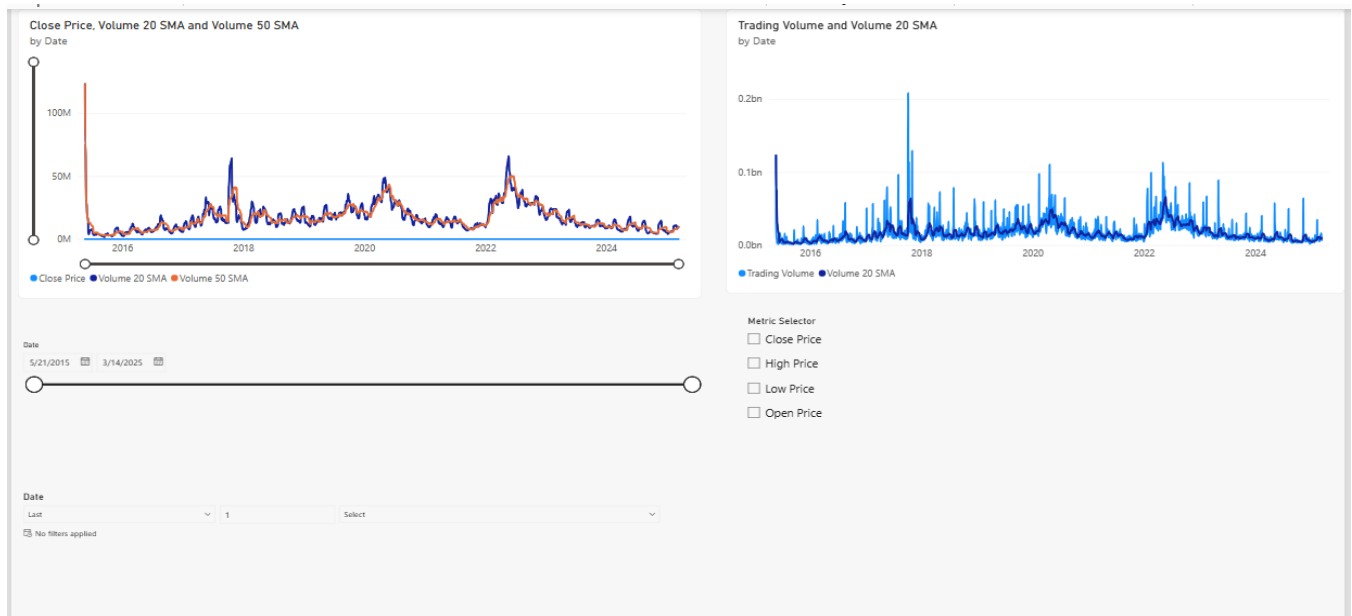
12. FINAL INTERACTIVE DASHBOARD :

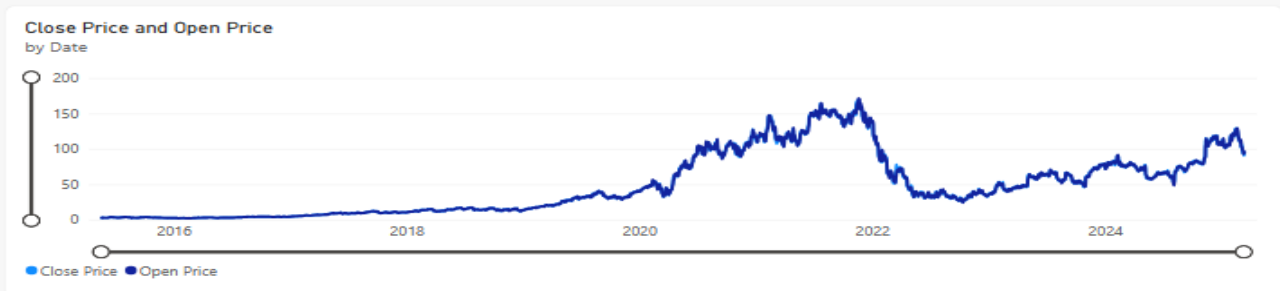
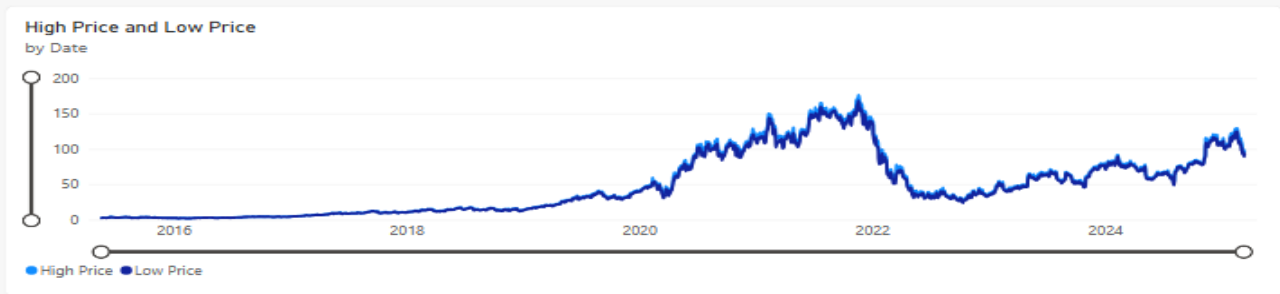
The final dashboard combines the major analytical components into a single Power BI report.

Dashboard Components

1. Price Trend & Moving Averages
2. Trading Volume & 20-Day SMA
3. Date Range Slicer
4. Relative Date Slicer
5. Metric Selector
6. High and Low Price Trend
7. Open and Close Price Trend
8. Custom Tooltip

The combination of these components provides an interactive view of Shopify stock price and trading-volume information.





Open Price
48.46

Trading Volume
41bn

Daily Return %
44.59

High Price
176.29

Low Price
1.85

Close Price
48.46

13. KEY ANALYTICAL FEATURES :

The dashboard provides multiple ways to explore the Shopify stock dataset.

Short-Term Trend

The 20-day SMA provides a shorter-period reference for trading volume.

Long-Term Trend

The 50-day SMA provides a longer-period comparison.

Moving Average Comparison

The crossover signal compares the 20-day and 50-day moving averages and returns either a bullish or bearish signal according to their relationship.

Price Comparison

High, Low, Open, and Close prices can be examined through separate visual comparisons.

Time-Based Exploration

The date slicers allow the analysis to be narrowed to selected or relative time periods.

Detailed Information

The custom tooltip provides additional stock information without requiring extra dashboard space.

14. KEY LEARNING OUTCOMES

Through this activity, I gained practical experience in several Power BI concepts.

DAX

I learned how to create measures using functions such as:

- VAR
- MAX ()
- DATESINPERIOD ()
- AVERAGE ()
- CALCULATE ()
- IF ()
- ISBLANK ()

Data Analysis

I gained experience in analysing:

- Stock prices
- Trading volume
- Moving averages
- Price comparisons
- Time-based trends

Power BI Features

I also learned how to work with:

- Date slicers
- Relative date filters
- Field Parameters
- Line charts
- Zoom sliders
- Custom tooltips
- Interactive dashboards

These concepts helped me understand how Power BI can be used to build an interactive analytical report rather than only displaying static charts.

15. CONCLUSION

The Week 4 Power BI project focused on performing Exploratory Data Analysis on Shopify stock data.

The analysis combined **DAX-based moving averages, stock price comparisons, trading-volume analysis, interactive date filtering, dynamic metric selection, and custom tooltip functionality.**

The 20-day and 50-day moving averages provided different time-based references, while the crossover measure enabled comparison between the two. The additional price and volume visualizations provided different perspectives for exploring stock behaviour.

Overall, this project helped me gain practical experience in **Power BI EDA, DAX calculations, interactive filtering, data visualization, and dashboard design**. The completed dashboard presents the Shopify stock information in an organized and interactive format.

Submitted By :

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